

MULTIMODAL SPEECH EMOTION RECOGNITION USING DEBERTA AND WAV2VEC2.0

Affective Computing Project

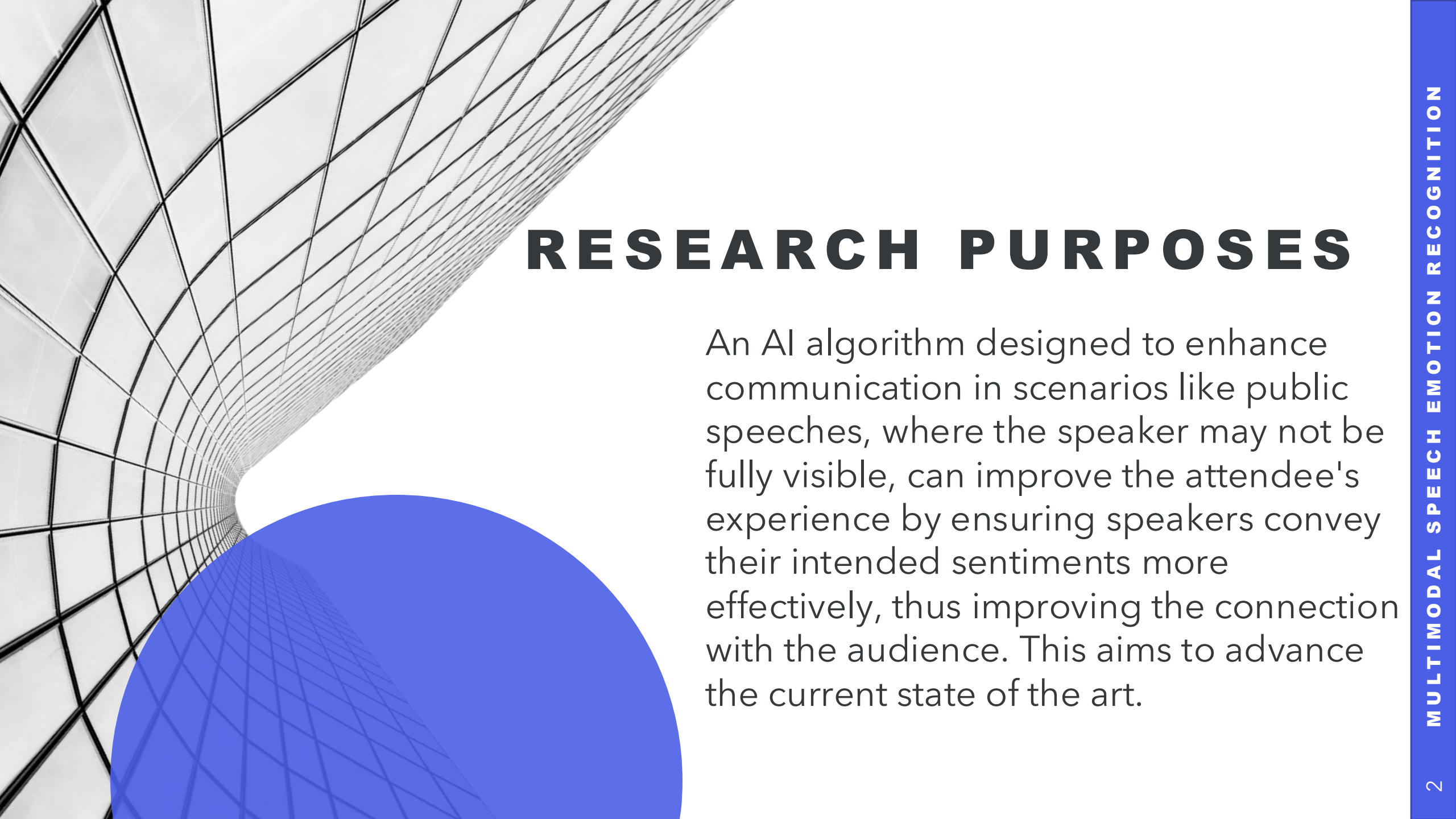
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RESEARCH PURPOSES

An AI algorithm designed to enhance communication in scenarios like public speeches, where the speaker may not be fully visible, can improve the attendee's experience by ensuring speakers convey their intended sentiments more effectively, thus improving the connection with the audience. This aims to advance the current state of the art.

METHODOLOGY

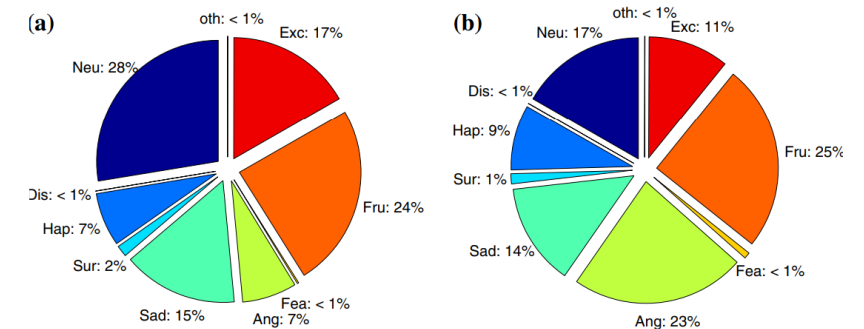
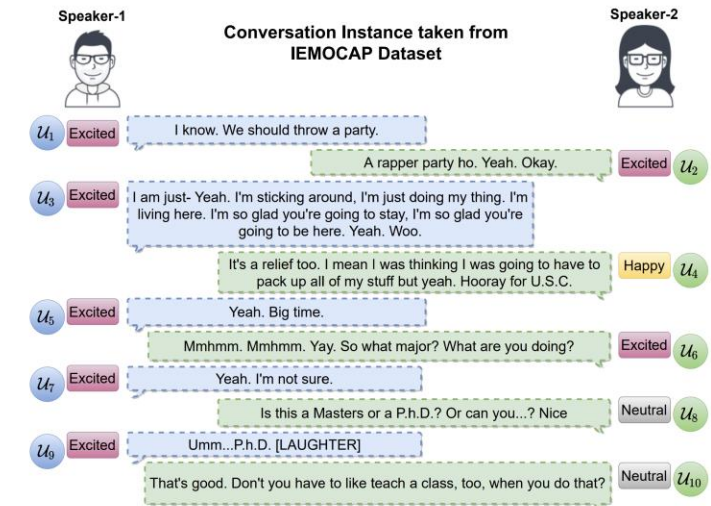


- What dataset did we use?
- How did we preprocess the data?
- Transfer learning techniques for DeBERTa and Vec2Wav2.0
- Designing the multimodal model
- How did we train the models and what metrics did we use?

IEMOCAP: Interactive EMotional Dyadic Motion CAPture Database

IEMOCAP is a multimodal and multi-speaker database that contains approximately 12 hours of audiovisual data, including video, speech, motion capture of the face, and text transcriptions. It is annotated into categorical labels: anger, excited, fear, sad, surprised, frustrated, happy, disappointed and neutral.

Due to the dataset's imbalance, evident in the right figure, we opted to focus on these emotions: Angry, Frustrated, Neutral, Sadness, Excited, and Happy. Furthermore, the dataset lacked a predefined split, leading us to divide it into 80% for training and 20% for validation. We also initialized all seeds to 0 to ensure result consistency.



```
INFO:scripts.run_model:Getting the train dataloader...
INFO:dataloaders.iemocap:Loaded train dataset. Size: 5984
INFO:dataloaders.iemocap:Each emotion percentages: ang - 977 | fru - 1507 | neu - 1345 | sad - 859 | exc - 764 | hap - 452
INFO:dataloaders.iemocap:Shapes: audio - (54622,); text - (537,)

INFO:scripts.run_model:Getting the eval dataloader...
INFO:dataloaders.iemocap:Loaded test dataset. Size: 1476
INFO:dataloaders.iemocap:Each emotion percentages: ang - 126 | fru - 342 | neu - 363 | sad - 225 | exc - 277 | hap - 143
INFO:dataloaders.iemocap:Shapes: audio - (54622,); text - (537,)
```

Data Preprocessing

To be able to quickly make our experiments, we preprocessed the data, so the dataloader wouldn't need to process all the values at runtime.

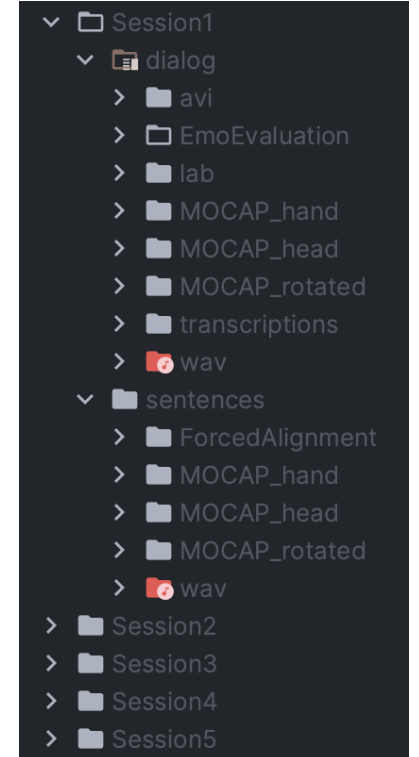
We processed the IEMOCAP dataset (compact, video-free version), which contains five session directories, each with 'dialog' and 'sentences' subdirectories. Using the 'EmoEvaluation' directory, we linked audio files to their respective emotions. Then, via the 'transcriptions' directory, we matched these files to the actors' spoken words, resulting in a compiled table of audio files, text, and emotions.

```
% [START_TIME - END_TIME] TURN_NAME EMOTION [V, A, D]

[4.8800 - 7.7400] Ses01F_impro03_F000 hap [4.3333, 2.6667, 2.6667]
C-E1: Happiness; ()
C-E2: Happiness; ()
C-E4: Happiness; ()
C-F1: Excited; Happiness; ()
A-E1: val 4; act 3; dom 3; ()
A-E2: val 5; act 3; dom 3; ()
A-E3: val 4; act 2; dom 2; ()
A-F1: val 5; act 4; dom 4; ()

[7.9300 - 11.3700] Ses01F_impro03_F001 hap [4.0000, 2.6667, 2.6667]
C-E1: Excited; ()
C-E2: Happiness; ()
C-E4: Happiness; ()
C-F1: Excited; Happiness; ()
A-E1: val 4; act 3; dom 3; ()
A-E2: val 4; act 3; dom 3; ()
A-E3: val 4; act 2; dom 2; ()
A-F1: val 5; act 4; dom 4; ()
```

```
Ses01F_impro01_F003 [019.2900-020.7875]: You did.
Ses01F_impro01_F004 [021.3257-024.7400]: You were standing at the beginning and you directed me.
Ses01F_impro01_M003 [023.4700-028.0300]: Okay. But I didn't tell you to get in this line if you are filling out this particular form.
Ses01F_impro01_F005 [027.4600-031.4900]: Well what's the problem? Let me change it.
Ses01F_impro01_M004 [028.3950-031.2117]: This form is a 7.X.four.
```



Data Preprocessing

Furthermore, we processed the raw audio files by passing the raw audio data to the Wav2Vec2 processor correspondent to the model we used (facebook/wav2vec2-base-960h), which wraps a Wav2Vec2 feature extractor and a Wav2Vec2 CTC tokenizer into a single processor and exported the table containing the output of the audio processor, the raw text data and the emotion label.

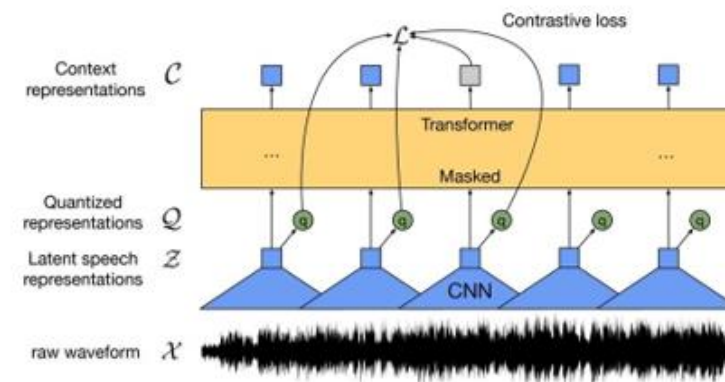
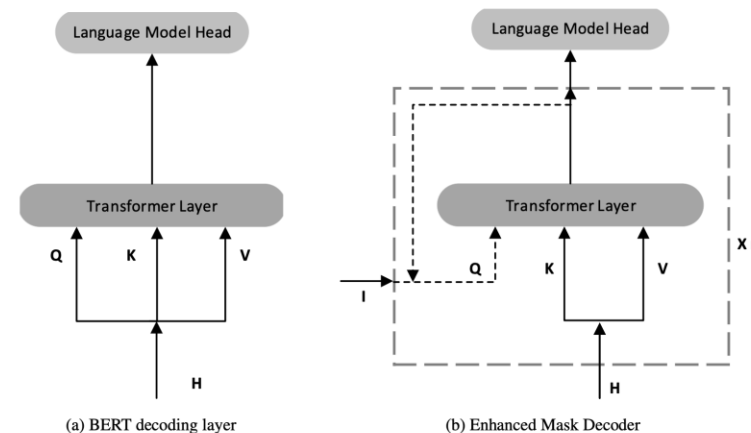
Lastly, we processed the raw text data by passing it to the DeBERTa AutoTokenizer, receiving some tokens, which we exported in a table along the processed audio data and the emotion label.

Because we wanted to train the model on mini-batches to improve generality, at these steps, we padded the processed audio and text data to match the one with the longest length. As a compromise, because some of the biggest audio data were way too big (546220), and we didn't have the required computing power, we truncated the data a 10th of its length, so some of the longer audio data wouldn't be very accurate, but most of them aren't in this category, so it shouldn't drastically affect our results.

Transfer Learning – audio and text

We utilized the DeBERTaV3-Small model by Microsoft, featuring 44M backbone parameters and a 128K token vocabulary, for text data analysis of IEMOCAP. Chosen for its superior performance, DeBERTa uses disentangled attention and an enhanced mask decoder, outperforming BERT and RoBERTa in Natural Language Understanding tasks with 80GB of training data.

For audio processing, we employed Facebook's Wav2Vec2-Base model, trained on 960 hours of 16kHz LibriSpeech audio and boasting 94.4M parameters. Known for its self-supervised learning approach and effectiveness in speech recognition, Wav2Vec2 excels in low-labeled data environments. The model's efficiency is highlighted by its performance on the 100-hour LibriSpeech benchmark, where it outshines previous models using just one hour of labeled data.



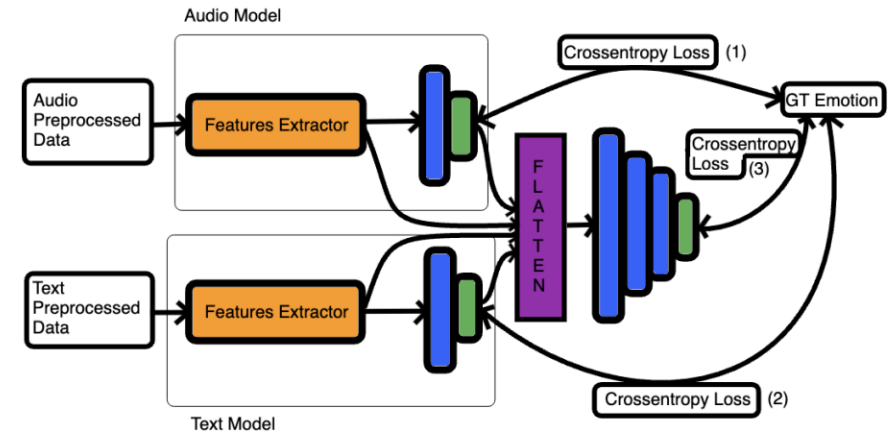
Designing the multimodal model

Initially, we fine-tuned the Wav2Vec2.0 audio model and the DeBERTa text model on the IEMOCAP dataset. For the subsequent training of the Fusion Model, we froze the weights of both the audio and text models. We then combined the features and classifications from both models, flattened this data, and processed it through a 4-layer Multi-Layer Perceptron (MLP). The optimization of the MLP's output was trained to match the emotions, ensuring that the only learnable weights in this stage were those of the MLP.

Training

We opted for ADAM (Adaptive Moment Estimation) as our optimizer due to its rapid convergence, enabling us to conduct more experiments. Although many papers favor SGD (Stochastic Gradient Descent) for its accuracy in convergence, it is notably slower.

For our loss function, we used cross-entropy loss, a common choice for classification tasks. While more complex options like CTC (Connectionist Temporal Classification Loss) exist, we found cross-entropy sufficient for our needs.

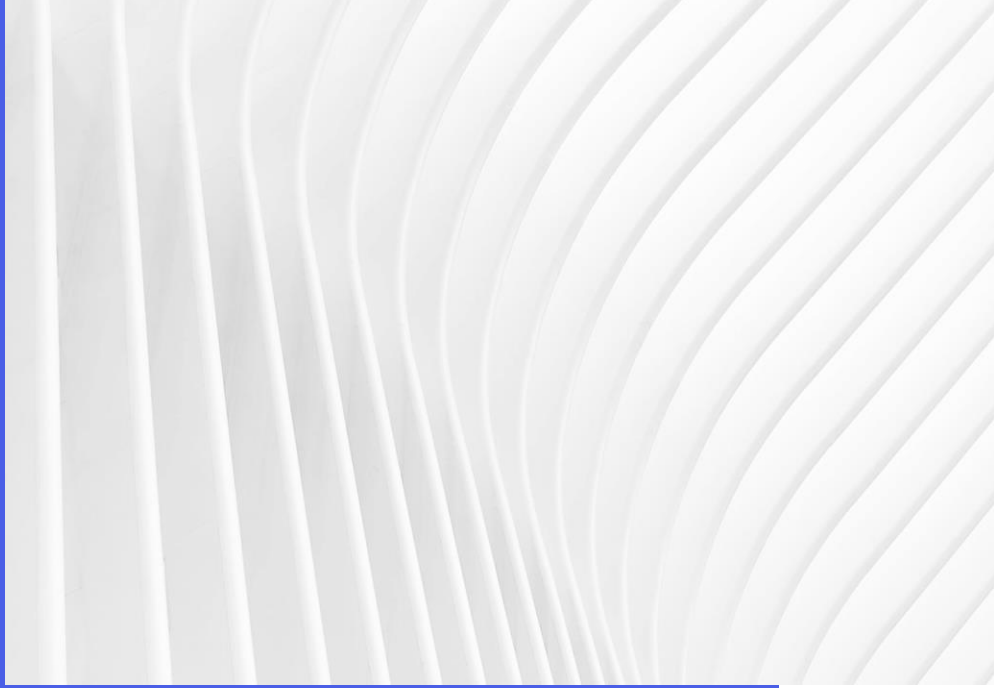


- $m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$
- $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$
- $\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$
- $\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$
- $\theta_t = \theta_{t-1} - \frac{\text{lr} \cdot \hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} - \text{lr} \cdot \text{weight_decay} \cdot \theta_{t-1}$

where:

- m_t is the first moment estimate.
- v_t is the second moment estimate.
- β_1 and β_2 are the exponential decay rates for the moment estimates.
- g_t is the gradient at time step t .
- $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected moment estimates.
- lr is the learning rate.
- ϵ is a small constant to prevent division by zero.
- weight_decay is the weight decay term.

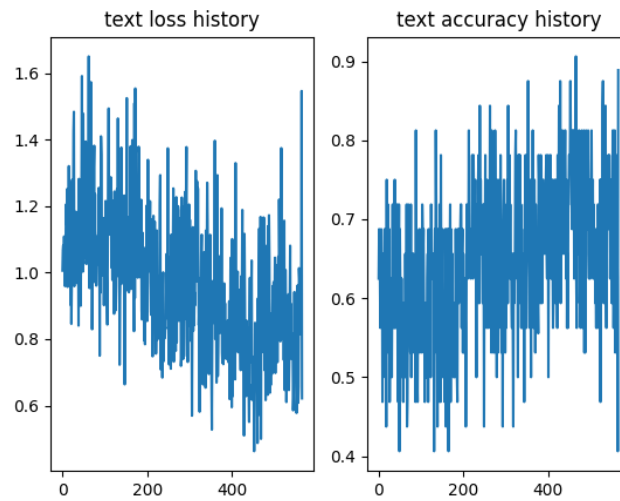
RESULTS



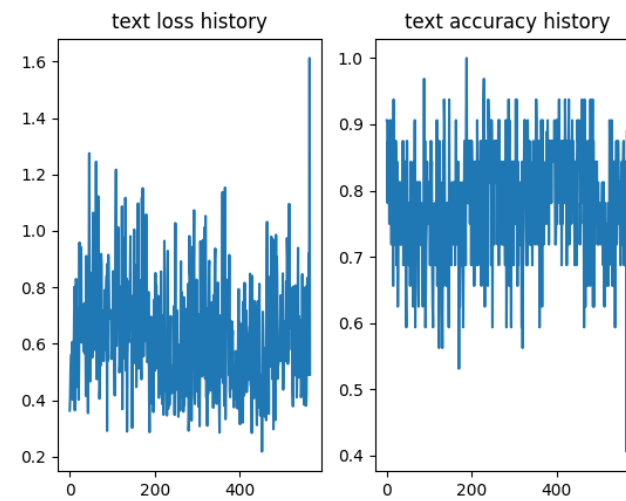
DeBERTa



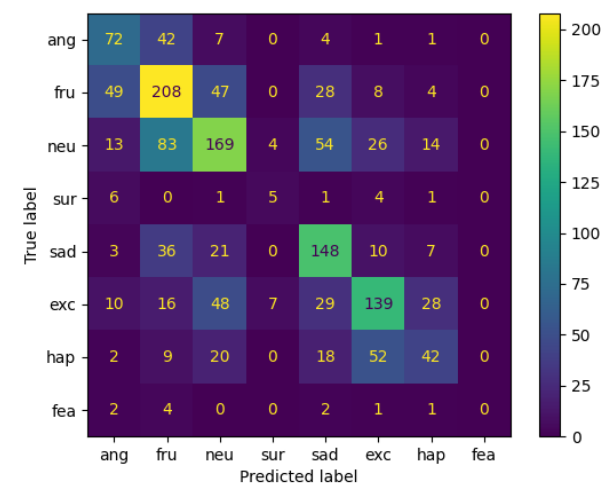
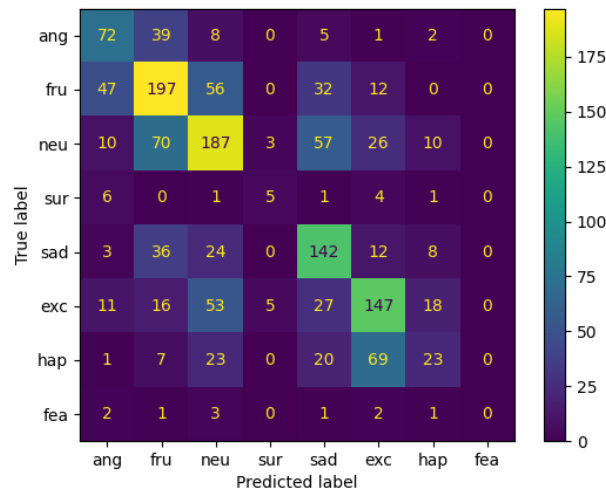
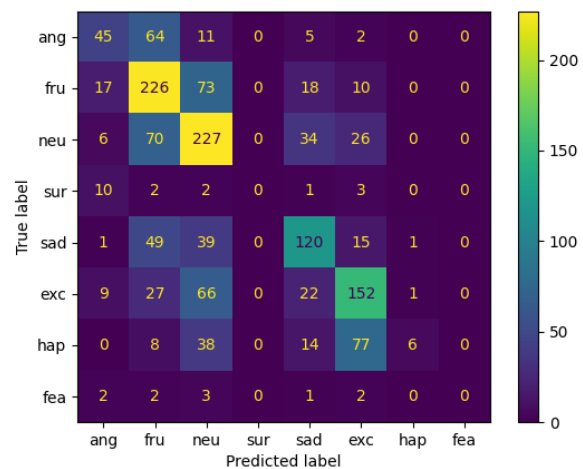
Epochs 1 -> 3



Epochs 4 -> 6



Epochs 7 -> 9



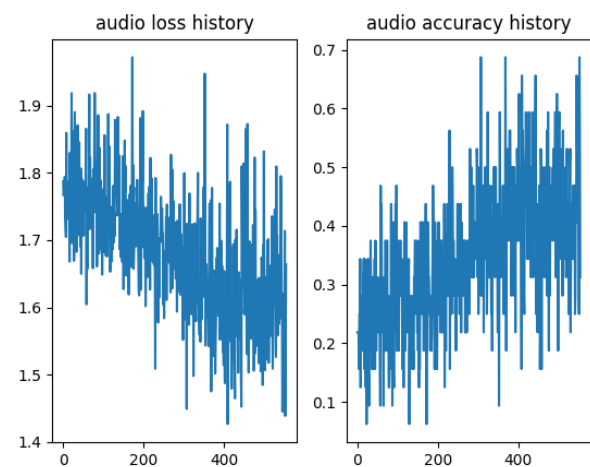
DeBERTa results interpretation

The model required approximately 45 minutes per epoch to train on a MacBook M2 Pro with 12 CPU cores, 19 GPU cores, and 32GB RAM. This was before we removed the three least used classes. Initially, the loss function showed good convergence, but later epochs indicated no further training was needed as the validation dataset's accuracy remained unchanged. This suggests possible overfitting. The confusion matrix revealed the model's difficulty in distinguishing between similar emotions, which could be contributing to its generalization challenges.

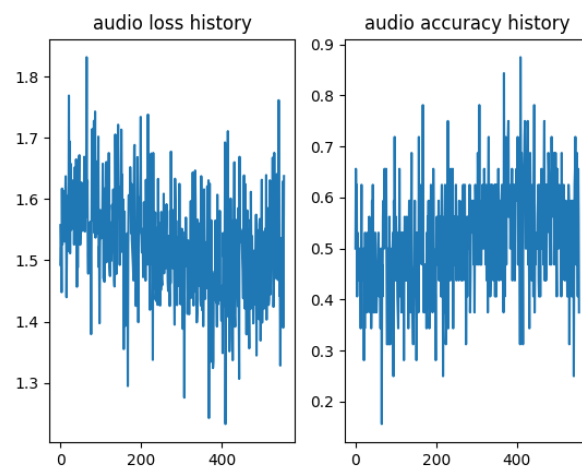
Epoch Start	Epoch End	Accuracy
1	3	51.493%
4	6	51.293%
7	9	51.957%

Model	Accuracy
bc-LSTM + Attention (2017)	58.540% (UA)
Attention-BLSTM (2018)	62.900% (UA)
Dialogue-CRN + RoBERTa (2021)	67.530% (UA)
EmoCAPS (2022)	71.770% (UA)
DeBERTa (ours)	51.957% (6 classes - UA)

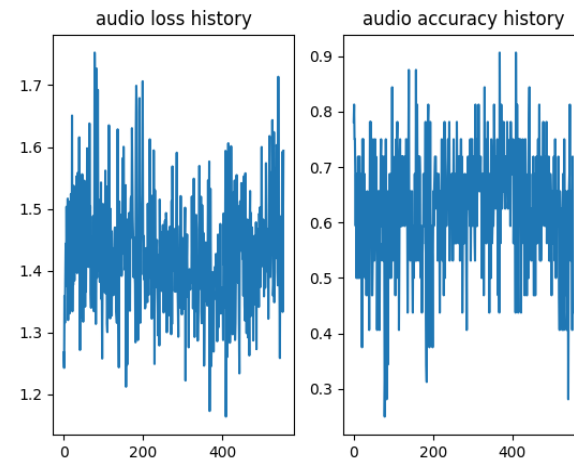
Wav2Vec2.0



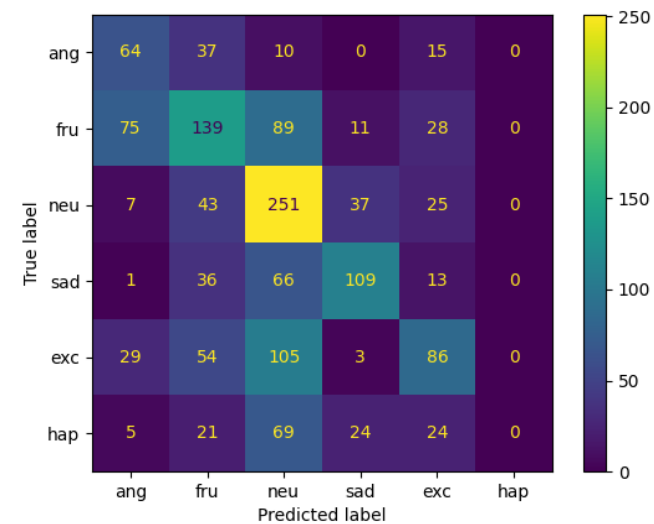
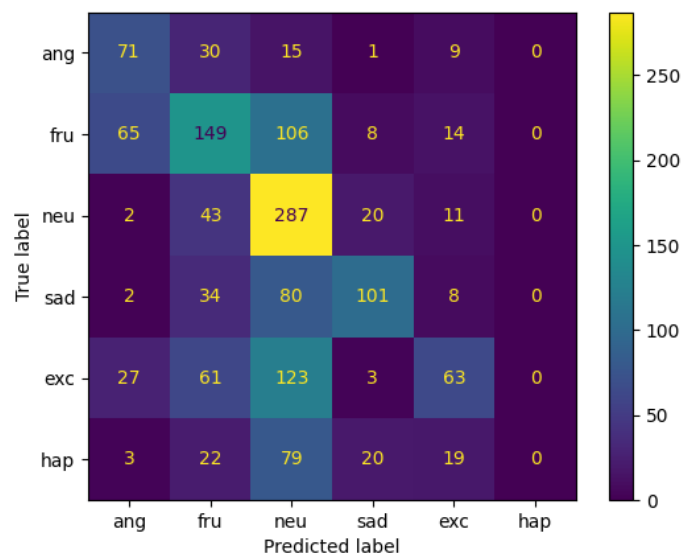
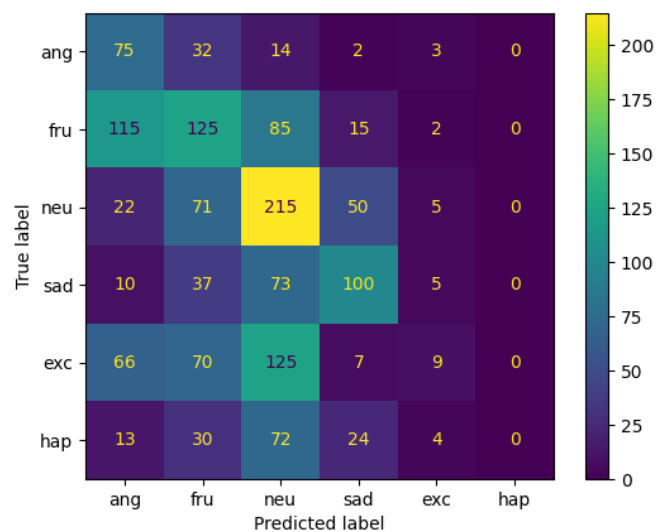
Epochs 1 -> 3



Epochs 4 -> 6



Epochs 7 -> 9



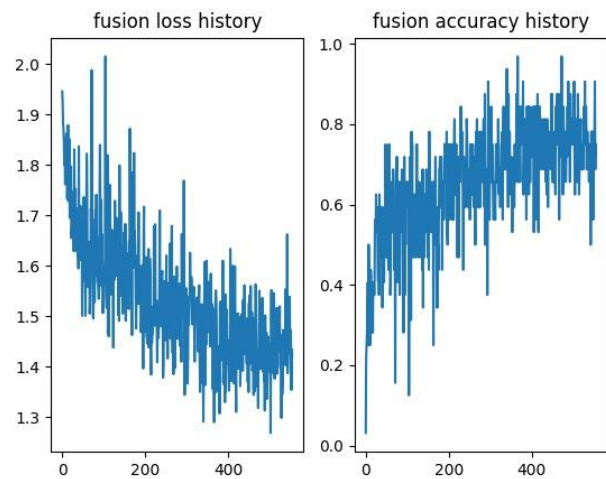
Wav2Vec2.0 results interpretation

The model's training time was about 25 minutes per epoch on the MacBook, similar to DeBERTa's. Considering we shortened the audio data to a tenth of its longest length, the results were satisfactory. The accuracy stabilized after 4-6 epochs, so further training was unnecessary. A notable issue in the confusion matrices was the model's reluctance to identify the 'happy' emotion, a phenomenon also observed in the DeBERTa results. Despite adding a mock class and extensive debugging, the reason for this bias remains unclear, as the model consistently assigned low values to the 'happy' label.

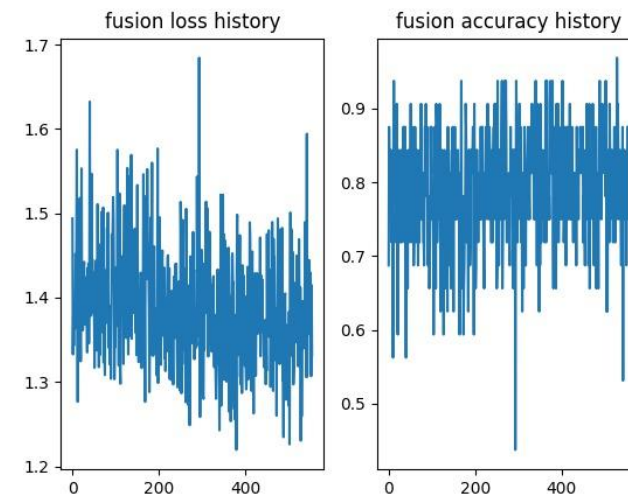
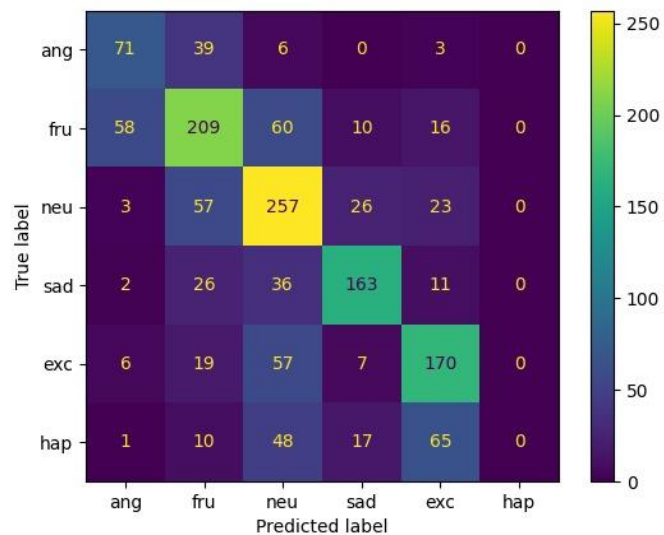
Epoch Start	Epoch End	Accuracy
1	3	35.501%
4	6	45.461%
7	9	43.970%

Model	Accuracy
CNN + LSTM (2018)	61.7% (UA)
SYSCOMB (2020)	74.0% (UA)
<i>SER with MTL (2021)</i>	78.9% (UA)
MMER (2022)	81.7% (UA)
Wav2Vec2.0 (ours)	45.461% (6 classes)

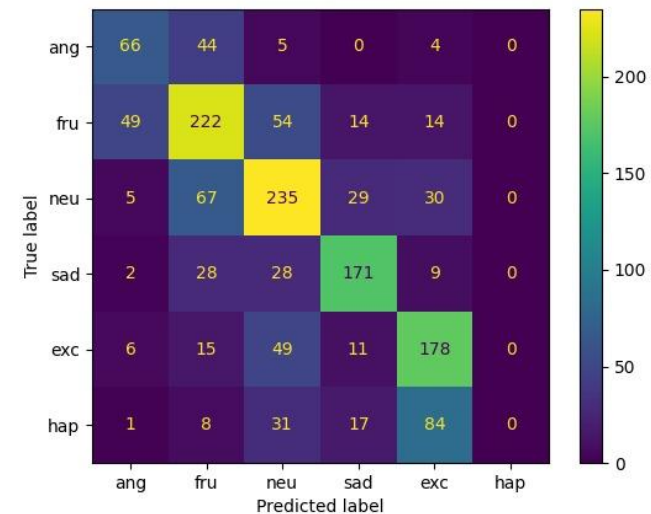
Fusion Model



Epochs 1 -> 3



Epochs 4 -> 6



Fusion Model results interpretation

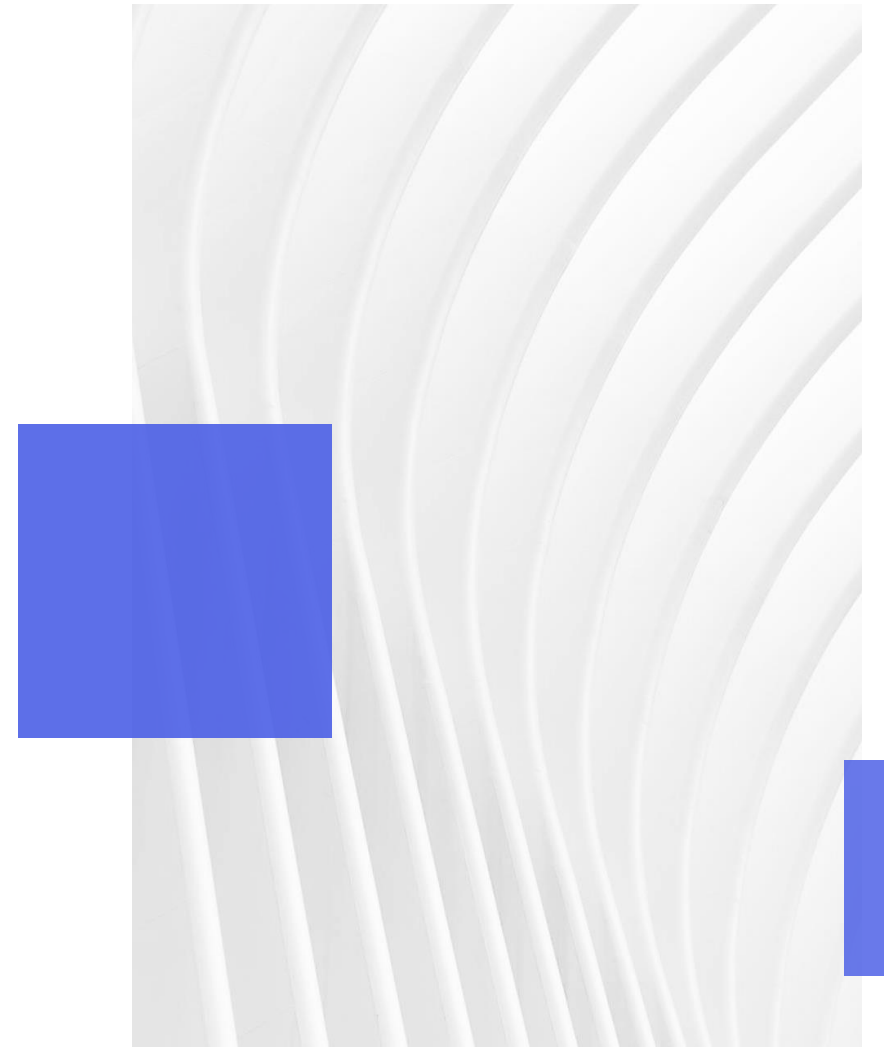
Training this model locally would have been time-intensive, so it was trained on a 16GiB P5000 GPU with 8 CPU cores and 30GB RAM. The results surpassed those of the individual models, likely due to the MLP's skill in merging data and resolving uncertainties from either model. However, the model consistently failed to identify the 'happy' emotion, an issue also noted in previous models, despite DeBERTa's occasional correct recognition of 'happy' data.

Epoch Start	Epoch End	Accuracy
1	3	58.943%
4	6	59.078%

Multi Model	Accuracy
CHFusion (2018)	75.9% (UA)
Self-attention weighted correction (2022)	76.8% (UA)
COGMEN (2022) (ranked first)	68.2% (UA - 6 Classes) 84.5% (F1, 4-class)
DeBERTA + W2V2.0 (ours)	59.078% (UA - 6 classes)

CONCLUSIONS AND FUTURE WORK

Currently, the project utilizes smaller versions of the DeBERTa and Wav2Vec2 models (DeBERTa small and Wav2Vec2 base). For future enhancements, we plan to implement their larger counterparts, anticipated to boost performance due to their increased complexity and capacity. Additionally, more extensive hyperparameter tuning will be conducted. This approach is expected to lead to improved outcomes in the project.





THANK YOU



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